INCH-POUND
MIL-PRF-3098/101E
w/AMENDMENT 1
16 August 2023

SUPERSEDING
MIL-PRF-3098/101E
1 November 2017

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR124/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 2.9 MHz to 3.8 MHz; fundamental; noncontrolled; antiresonance.

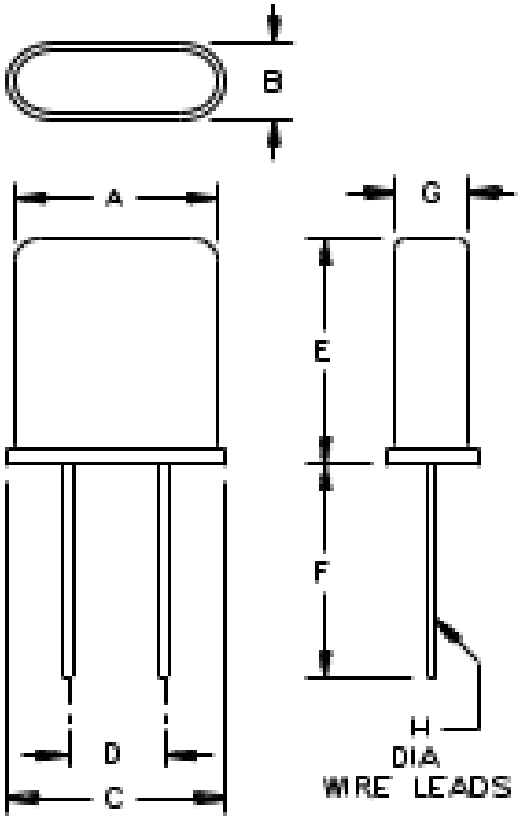



FIGURE 1. Crystal unit - CR124/U.

AMSC N/A
FSC 5955

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.

| Ltr | inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .402 | --- | 10.21 |
| B | --- | .183 | --- | 4.65 |
| C | --- | .435 | --- | 11.05 |
| D | .184 | .200 | 4.67 | 5.08 |
| E | --- | .530 | --- | 13.46 |
| F | .50 | --- | 12.7 | --- |
| G | --- | .150 | --- | 3.81 |
| H | .015 | .019 | 0.38 | 0.48 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR124/U, crystal unit - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.9 MHz to 3.8 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance:

Operating temperature range: ± 50 ppm.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Resistance: 150 ohms.

Crystal current: 2.7 mA.

Antiresonance, load capacitance: 32.0 pF ±.05 pF.

Operating temperature range (noncontrolled): -55°C to +100°C, inclusive.

Operable temperature range: -65°C to -55°C and +100°C to +105°C, inclusive.

Equivalent resistance: 450 ohms, maximum.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: MIL-STD-202-204, test condition A; 0.10 inch double amplitude or 5 gravity units (55 cps to 500 cps), whichever is less.

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: ± 15 percent.

Thermal shock: ± 50 ppm.

Aging:

Frequency change permitted: ± 5 ppm.

Qualification: Not required. First article testing (FAT) shall be performed according to the qualification table II of MIL-PRF-3098; for product level B, Table II is applicable and for product level S, Table II-A is applicable. Thirty samples for product level B and forty-five samples for product level S shall be selected. Approval of FAT shall be at the discretion of the acquiring activity..



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Air Force - 19

Preparing activity:
DLA – CC
(Project 5955-2023-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:49Z
Check the source to verify that this is the current version before use.